

Prevalence and initiation of preferential flow paths in a sandy loam with argillic horizon

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Abstract

Numerous studies have reported on preferential solute transport, giving evidence that preferential flow is widespread. However there has been little field documentation of the relative importance of different preferential transport mechanisms. This study used a dye tracer to examine the extent and relative importance of different preferential transport mechanisms in a glacial outwash-derived soil that is used extensively for high input agriculture in central Minnesota, USA (Verndale sandy loam: coarse loamy over sandy, mixed, frigid, Udic Argiborolls). Experimental treatments included three initial soil water contents (WET, MEDIUM, and DRY) and three dye solution application rates (FLOOD, SPRINKLER-High, and SPRINKLER-Low). Thirteen cm of FD&C Blue no. 1 (also known as Brilliant Blue FCF) food dye solution (200 g l^{-1}) were applied to replicated $1 \text{ m} \times 1 \text{ m}$ plots in a recently tilled 5 year old alfalfa stand and to two additional plots with no history of alfalfa. Vertical soil profile faces were exposed at 10-cm increments across each plot. Extensive and deeper preferential dye movement occurred under FLOOD conditions regardless of initial soil moisture or recent vegetation history. The two SPRINKLER rates generally resulted in relatively shorter preferential flow paths (PFPs). Within-plot variability of dye patterns—including depth and number of PFPs—was very high. Most PFPs observed were associated with roots and decayed roots, or with patterns in the abruptness and topography of the boundary between the Ap and Bt horizons. Open burrows were uncommon, but contributed to extensive preferential flow in the two NO-ALFALFA plots. Our findings indicate that preferential transport is prevalent under the variety of application rate and soil moisture conditions evaluated, and that observable soil features appear to be initiators of the majority of the PFPs. Only a few (10

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of 126) of the profiles excavated had preferential flow paths that were not associated with visible soil features. The observed high variability gives support to the idea that observations of spatial variability in pesticide transport studies is due to preferential transport. Our initial goal of elucidating the relationship between rate of application and the relative number and depth of PFPs was aimed at evaluating *water* flow patterns. However, subsequent research found that dye retardation was increased at slower application rates, indicating that the dye patterns we observed were due to both the rate and the pattern of water movement. As a result, we caution that our findings of generally deeper and more extensive preferential dye transport under the higher velocity FLOOD application rate do not necessarily indicate more extensive preferential water (or non-adsorbing solute) transport at this rate compared to the intermittent SPRINKLER rate. It is possible that our observations may be indicative of patterns in movement of adsorbed solutes such as pesticides, however this contention would require further research. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

Numerous studies have reported the occurrence of preferential solute transport, giving evidence that preferential flow is widespread (Jury and Roth, 1990). Preferential transport of solute and water through soil leads to accelerated transport of chemicals below the root zone—that is, faster movement than would occur with a uniform wetting front.

Field studies have demonstrated the occurrence of preferential movement of water or solutes through macropores and cracks (e.g., Beven and Germann, 1982), as a result of soil layering (Kung, 1990a; van Wesenbeeck and Kachanoski, 1994), and as a result of hydrophobicity (e.g., Hendrickx et al., 1993; Ritsema et al., 1993). Other field studies have documented the occurrence of preferential transport, without attempting to discern the responsible mechanism(s) (e.g., Ghodrati and Jury, 1990, 1992; Flury et al., 1994), or have set out to observe the occurrence of a specific mechanism (e.g., Starr et al., 1978, 1986; Glass et al., 1988; Edwards et al., 1993). Numerous theoretical and laboratory analyses have investigated different mechanisms for preferential flow including macropore flow (e.g., Booltink et al., 1993; Munyankusi et al., 1993; Li and Ghodrati, 1994; Buttle and Leigh, 1997), unstable flow (e.g., Hill and Parlange, 1972; Raats, 1973; Philip, 1975; Hillel and Baker, 1988; Glass et al., 1989; Selker et al., 1992), and funnel flow (Kung, 1990b; Ju and Kung, 1993). Few studies, however, have been designed to evaluate the relative importance these different mechanisms play in a given field situation.

Understanding the relative roles played by different preferential transport mechanisms is important for predicting when it will occur, and the relative acceleration of transport expected. Existing methods for predicting chemical transport in preferential flow situations emphasize prediction of transport under

a specific mechanism such as macropore flow (Chen and Wagenet, 1992; Booltink et al., 1993) or funnel flow (Ju and Kung, 1993), or are stochastic (van der Zee and Boesten, 1991; Roth and Jury, 1993).

In this work, our interest was to examine the extent of preferential transport and the relative importance of different preferential transport mechanisms in the Verndale sandy loam. This is a glacial outwash-derived soil that is used extensively for high chemical input, often irrigated, agriculture, and is found in a region of declining drinking water quality. Research with nitrate movement in this soil has shown that the greatest risk of groundwater contamination by agricultural chemicals often occurs when a rainfall comes soon after an irrigation event (Sexton et al., 1996)—i.e., when the soil is already ‘wet’. The Verndale sandy loam has a variety of characteristics (Table 1) that could allow preferential flow through one or more of the above mechanisms. It has several contrasts in hydraulic conductivity (between horizons in the upper profile and between depositional layers lower in the profile) that could promote layer-induced preferential flow (e.g., funnel flow, fingering). Macropore flow could also be important—especially as a result of root channels and faunal activities. Interpedal cracks and earthworm burrows were not expected to be important contributors to preferential flow in this soil, since there is only weak to moderate structure and the abrasiveness of the sand is not conducive to the presence of earthworms; however ant burrows were numerous.

The objectives of our research were:

1. To determine the prevalence of preferential flow and the depth of dye penetration under a variety of initial moisture contents and solution application rates (i.e., time periods over which dye solution is applied);
2. To determine the extent to which observed preferential flow paths (PFPs) were associated with the various observable morphologic features discussed above.

To meet these objectives, we conducted a field study consisting of applying a solution of dyed water to replicated plots under a variety of solution application rates and a variety of initial moisture contents. The variety of application rates and initial moisture contents was thought to be important because different preferential flow mechanisms are expected to occur under different conditions. Excavation of a series of vertical faces in each plot allowed evaluation of the prevalence and characteristics of PFPs, including association with observable morphologic features in each profile.

Since conducting the field work and analyses described in this study, we learned of a similar study conducted independently by Flury et al. (1994) using a similar approach. The similarities between studies allow some comparison of the results, while the differences between the studies allow us to enlarge the scope of current knowledge, particularly giving us detailed insight into the characteristics of preferential flow in one particular soil in which a variety of preferential flow mechanisms have potential to occur.

Table 1
General properties of the Verndale sandy loam

Horizon	Depth cm	Sand	Silt	Clay	Organic carbon	Soil structure ^a	Bulk density Mg m ⁻³	pH 1:1 CaCl ₂	Water content		Sat. hydraulic conductivity. mm h ⁻¹
		kg kg ⁻¹							- 0.01 MPa	- 1.5 MPa	
Ap	0–26	0.79	0.13	0.08	0.0124	w,c,gr	1.55	5.3	0.23	0.07	47.4
Bt	27–40	0.71	0.19	0.10	0.0050	m,f,sbk	1.55	5.4	0.26	0.10	13.8
BC	41–52	0.80	0.13	0.07	0.0009	w,m,sbk	1.48	–	–	–	49.8
C	53–100	0.92	0.05	0.03	0.0002	single gr	1.53	5.7	0.13 ^b	< 0.04 ^c	120 ^b

Soil organic carbon, bulk density and pH were measured on subsamples of soil collected from the field site described in this study. Unless otherwise noted, all other values were measured by Sexton (1993) on the same soil type about 200 m away.

^aGrade: w = weak, m = moderate; size: c = coarse, f = fine, m = medium; type: gr = granular, sbk = subangular blocky, single gr = single grain.

^bFrom Pang (1995).

^cWater content at – 1.2 MPa was 0.04 m³ m⁻³ (Pang, 1995).

2. Methods and materials

2.1. Experimental site

This experiment was conducted in August 1993 at the Central Minnesota Economic Development Research and Education Center in central Minnesota (USA) near the town of Staples (approximate latitude, longitude of 46°25', 94°45'). The field chosen for the experiment had a five year old stand of alfalfa. Two additional plots were studied in 1994 about 100 m distant in an uncropped area that had no history of alfalfa. The soil is a Udic Argiboroll (Verndale sandy loam), formed in sandy glacial outwash (Table 1). It consists of an 18–26 cm thick sandy loam Ap horizon underlain by a sandy loam argillic horizon that is about 14 cm thick. Soil structure was not observable while in place, however upon removal from the profile, the Ap horizon (below the rototilled zone) had a weak granular structure, and the Bt horizon had fine, moderate subangular blocky structure. Sandier 2BC and 2C horizons lie below the Bt horizon. Typically this soil is used for irrigated potato, corn, and/or kidney bean production. Slopes are minimal (0–2%).

2.2. Dye characteristics

The dye we used was FD&C Blue #1 (food grade, shipped as water-soluble powder) obtained from the Warner–Jenkinson, St. Louis, MN.¹ This dye is also known as C.I. Food Blue 2, and as Brilliant Blue FCF (Disodium salt of ethyl [4-[*p*[ethyl (*m*-sulfobenzyl)amino]-*a*-(*o*-sulfophenyl)benzylidene]-2,5-cyclohexadien-1-ylidene] (*m*-sulfobenzyl) ammonium hydroxide inner salt). This dye was chosen based on good contrast with our soil, low toxicity implied by its food dye designation, and results of preliminary studies using repacked soil columns and ponded conditions that showed faster movement through the Ap horizon of our field soil compared to Rhodamine WT. Lot analysis of the dye by the company listed the dye as 89% pure dye, with moisture, salt, and subsidiary dyes making up the rest. FD&C Blue #1 is non-fluorescent and has pKa's of 5.8 and 6.58, meaning two of its functional groups act as anions above these pH values. The pKa's plus additional dye characteristics are presented by Flury and Flühler (1994, 1995). Linearized Freundlich distribution coefficients (K_d) determined for the Ap, Bt, and C horizons at the study site are 4.33, 23.95, and 4.59 l kg⁻¹ respectively (Perillo et al., 1998).

¹ Identification of company or brand names do not constitute endorsement, but are included to provide a more descriptive listing of materials.

2.3. Plot setup

Experimental treatments consisted of three application rates (FLOOD, SPRINKLER-HIGH, and SPRINKLER-LOW), and three initial soil moisture conditions (WET, MEDIUM, and DRY). The specifics on these treatments are given below. FLOOD application was applied under each of the three initial moisture conditions, whereas the two SPRINKLER application treatments were applied to the WET and DRY initial conditions. Plots were 1 m $\times$ 1 m, set 1 m apart, in a randomized complete block design with two replications. We considered replication important to see if the patterns observed with respect to imposed treatments were consistent, or if soil heterogeneity outweighed observed patterns. Initial soil moisture was the main plot and application rate was the split plot. A third replicate was set up, but not evaluated. Due to logistics, neither of the DRY-SPRINKLER plots in the second replication could be evaluated either.

Prior to plot installation, each of the plot areas was rototilled to 8 cm depth in order to disrupt macropore continuity with the soil surface. Plastic lawn-edging was used for plot borders, and was installed several centimeters into the soil. Initial moisture contents were established with pre-irrigation (using sprinklers). WET and MEDIUM plots received at least 130 mm pre-irrigation. The use of rain gages throughout the plot areas showed that there was some non-uniformity in the amount of water applied to the different plot areas due to windy conditions typical of the area. No runoff was observed. DRY treatments received no pre-irrigation, and were covered during the pre-irrigation to avoid potential drift from adjacent WET and MEDIUM plot areas. Assuming uniform, piston-like flow (no dispersion), 130 mm water would penetrate to about 55 cm with all depths above 55 cm at 'field capacity' water content. The difference between WET and MEDIUM is that the MEDIUM plots were allowed to drain for 2 days prior to dye solution application. Gravimetric moisture samples were taken between plots at the start of dye application, at depths of 0–15, 15–30, 30–60, and 60–90 cm, in order to determine initial water content for each treatment.

2.4. Dye application

Dye application consisted of applying 130 mm dye solution (200 g l⁻¹) at different application rates (i.e., over different time periods). FLOOD plots were instantaneously ponded and maintained at a hydraulic head of about 4 cm until all dye solution was added. It took less than 1 h for all of the dye solution to infiltrate into these plots. Burlap cloth was placed on the soil surface to minimize surface disturbance during dye additions. The SPRINKLER-HIGH and SPRINKLER-LOW treatments were designed to simulate unsaturated water flow conditions. Dye solution was applied with a hand-held spray wand connected to a constant pressure (413 kPa) regulator on a field sprayer with

large mixing tanks typically used for pesticide application. SPRINKLER-HIGH plots received four 14-s applications each hour for two days, while SPRINKLER-LOW plots received two 14-s applications each hour over four days. Individual application passes were alternated between north–south, then east–west directions. After each application, plots were covered with plastic tarps to prevent evaporation. Some localized ponding was observed on the soil surface at the SPRINKLER-HIGH application rate. No applications were made at night due to difficulty seeing the plots and spray wand.

Expressed on an hourly basis, the application rates were about 130 mm h^{-1} for FLOOD treatments, 7.6 mm h^{-1} for SPRINKLER-High, and 3.8 mm h^{-1} for SPRINKLER-LOW. The three application rates correspond to 940%, 55%, and 28% of the lowest saturated hydraulic conductivity in the soil profile (Bt horizon). In comparison, the 100-year 1-h and 24-h storm events in the study area are 76 and 140 mm respectively. Relatively high-intensity, short-duration storms are common in the summer, with a 2-year 1-h storm being 33 mm. When dye application was complete, large plastic tarps were laid over the entire plot areas (in addition to the small plastic covers over individual plots) until they could be excavated.

2.5. *Plots without alfalfa*

The procedure described above was repeated at two additional plots in late summer 1994 at an adjacent uncropped location of the same soil type that had no history of alfalfa within at least ten years. These plots allowed comparison with dye patterns in the same soil but with a different rooting history. Since the majority of the deep preferential flow paths observed in the 1993 plots (alfalfa plots) occurred in the FLOOD treatments, only FLOOD treatments were imposed in the no-alfalfa plots. All experimental procedures were the same as described above for the Fall 1993 plots, except that there was no pre-irrigation.

2.6. *Excavation of plots*

Nine successive vertical profile faces at 10-cm intervals were examined in each plot, beginning 10 cm into the plot (Fig. 1). Although profile faces at 0 and 100 cm were also examined they are not included in the analysis of the dye patterns because they represent the boundaries of the treated area. Procedures for examination of each profile face consisted of digging back the specified distance and preparing a smooth, clean vertical face using flat, pointed trowels. Each face was photographed, and notes were made in the field of the depth to horizon boundaries, characteristics of the horizon boundaries, maximum depth of the most significant (deepest) dyed zones, and observable features that appeared to be related to the observed preferential dye paths.

We used the following procedures to establish true vertical conditions and maintain a consistent coordinate system for each profile face examined. Prior to

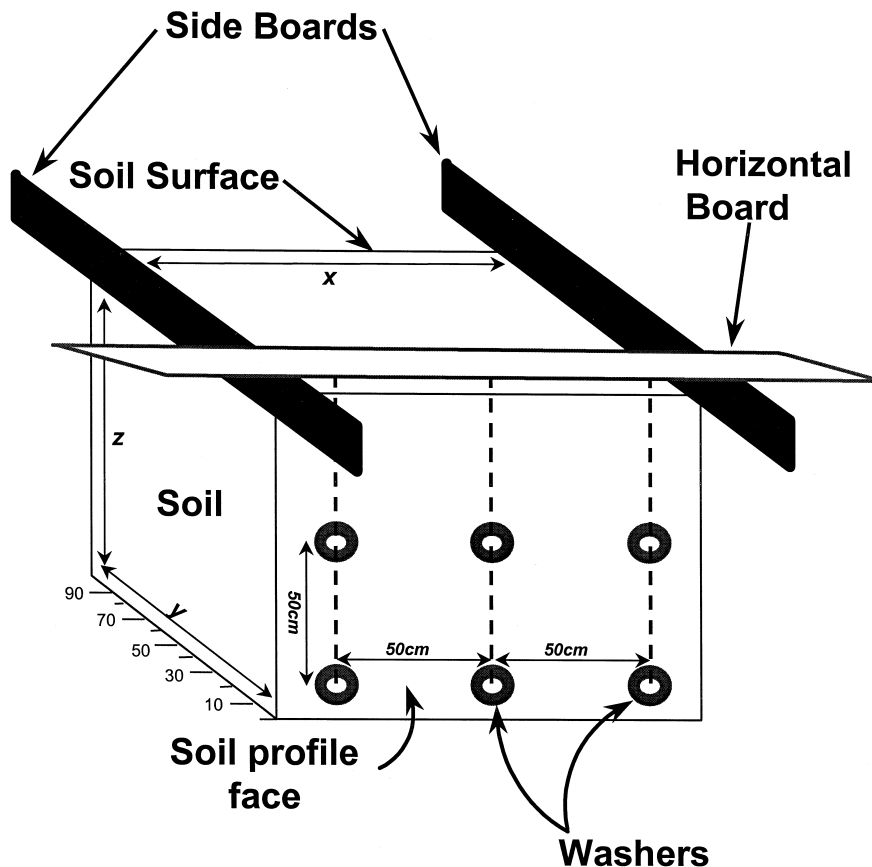


Fig. 1. Individual plot setup, showing board-and-washer arrangement used to maintain spatial coordinate system and to ensure true vertical profiles. Two leveled wooden boards were placed parallel to the plot, with a horizontal board laid across these to provide a zero coordinate in the vertical direction. Six metal washers (3.5 cm diameter) suspended from this board against the profile face denoted a true vertical face when they were plumb. The washers marked the 0, 50, and 100 cm coordinates across the face, and 50 and 100 cm below the zero coordinate in the vertical direction. The dashed lines in the drawing show the nylon line that was used to suspend the washers.

plot excavation, construction-grade wooden boards (2 cm by 15 cm) were installed in the soil surface outside of the plots parallel to the left and right sides of the face. The top edge of each board extended about 2.5 cm above the soil surface (Fig. 1), with each board leveled both to itself and with respect to each other. The top of the horizontal board was designated coordinate $z = 0$ (vertical direction), the left edge of the plot was designated $x = 0$ (left-to-right direction), and the front of the plot was designated $y = 0$ (distance into the plot). In order to record coordinates on the photographs and to facilitate excavating into the plot to exactly the distance desired, six 3.5-cm diameter metal washers (with orange centers) were hung from the top of the board using nylon line (Fig. 1).

The washers extended 50 and 100 cm below the top of the board at $x = 0$, 50, and 100 cm across the face. When the washers just barely hung freely we knew that the face was truly vertical. In this way, after correction for the height of the $z = 0$ coordinate above the soil surface, accurate three-dimensional coordinates were established for the entire plot and recorded on the photographs. The washers served as ‘tie points’ for determining location coordinates later in the analysis.

Color slide photographs (Ektachrome daylight) were taken with a tripod-mounted 35 mm camera placed at least 3.5 m from the first profile face (‘0’ profile). The same camera position was maintained for all profiles within a given plot in order to minimize distortion differences between successive profiles faces. Photographs were taken both in a naturally dry state as well as after misting the profile face with water using a plant mister. Photos taken in the dry state gave a good record of soil horization, since the sandier horizons quickly dried and lightened in color. Photos taken after misting allowed a better record of moist soil colors. The outlines of the dyed areas were digitized into ARC/INFO ver. 6.1, and then registered to a common coordinate system using the location coordinates discussed above. Following registration, the areas and vertical length of each dyed polygon were calculated, using a resolution of 5 mm. Our primary objective was to evaluate deep movement of the dye under the various treatments. In addition, it was sometimes difficult to discern dye patterns in the dark Ap horizon. For these reasons, only characteristics of dye patterns extending more than 25 cm below the soil surface were quantified.

2.7. Supporting measurements

During the examination of several profile faces, paired dye and no-dye cores (4.95 cm dia. by 5.1 cm long) were collected in order to determine soil water content and bulk density. Infiltration measurements were also made at both the alfalfa and no-alfalfa sites using double ring (+45 mm head) and tension infiltrometers (–35 mm head). Each method was replicated three times at each location, for a total of 12 measurements. Prior to infiltration measurement, the areas were rototilled to a depth of approximately 8 cm to obtain surface conditions similar to those used in the dye experiments. Additional details of plot setup and excavation are described in Perillo (1995).

3. Results and discussion

3.1. Soil characteristics

As expected, infiltration rates under saturated conditions (360 and 97.0 mm h^{-1} for alfalfa and no-alfalfa plots respectively) were significantly greater

($p < 0.01$) than under unsaturated conditions (139 and 66.3 mm h⁻¹ for alfalfa and no-alfalfa plots respectively). As these numbers show, infiltration rates were substantially higher in alfalfa plots compared to no-alfalfa plots, suggesting that alfalfa roots increased the rate of water movement. This is similar to the observation of Mitchell et al. (1995) who observed that decaying alfalfa tap roots formed stable macropores and played important roles in water movement. The MEDIUM and DRY initial water contents were not significantly different from each other (Table 2), though both were significantly ($p < 0.05$) lower than the WET initial water content condition. The lack of difference between water contents in the MEDIUM and DRY treatments indicates that most drainage occurred within two days after the large water additions at the surface. Throughout this manuscript the labels 'MEDIUM' and 'DRY' are kept for plot identification purposes.

3.2. Descriptions of preferential flow paths (PFPs)

In general, dye staining extended only 1 or 2 cm horizontally beyond the plot area, indicating that the majority of infiltrating solution moved downward into the plot area. As can be seen in Figs. 2 and 3, the dye front was generally not uniform in any of the experimental plots. The most dramatic preferential flow paths (PFPs) developed under FLOOD conditions. An example of the general shapes of dye paths observed, and the variability between nine profile faces separated by 10 cm in a single plot, is shown in Fig. 2 for a FLOOD treatment. The PFPs were generally long, column-shaped features with slight rounding at the bottom. An example of the patterns and variability among the nine profiles from one SPRINKLER plot is shown in Fig. 3. In general, both SPRINKLER-HIGH and SPRINKLER-LOW application rates resulted in relatively more

Table 2
Soil water contents immediately prior to dye application

Depth (cm)	Volumetric water content (m ³ m ⁻³)			
	Alfalfa plots			No-alfalfa plots ^b
	Wet	Medium	Dry	
0–15	0.265a ^a	0.227a	0.219a	0.081 (0.003)
15–30	0.232a	0.192b	0.206b	0.102 (0.007)
30–60	0.262a	0.184b	0.164b	0.091 (0.006)
60–90	0.179a	0.082b	0.116b	0.050 (0.006)
Total water in 90 cm profile (cm)	20.7	14.3	14.8	7.0

Each value is the mean of three measurements for 'Alfalfa' plots.

^aValues followed by the same letter in a given row indicate no significant differences between treatments at $p < 0.05$.

^bValues for 'No-alfalfa' plots are the mean of 12 measurements, followed by standard deviations in parentheses.

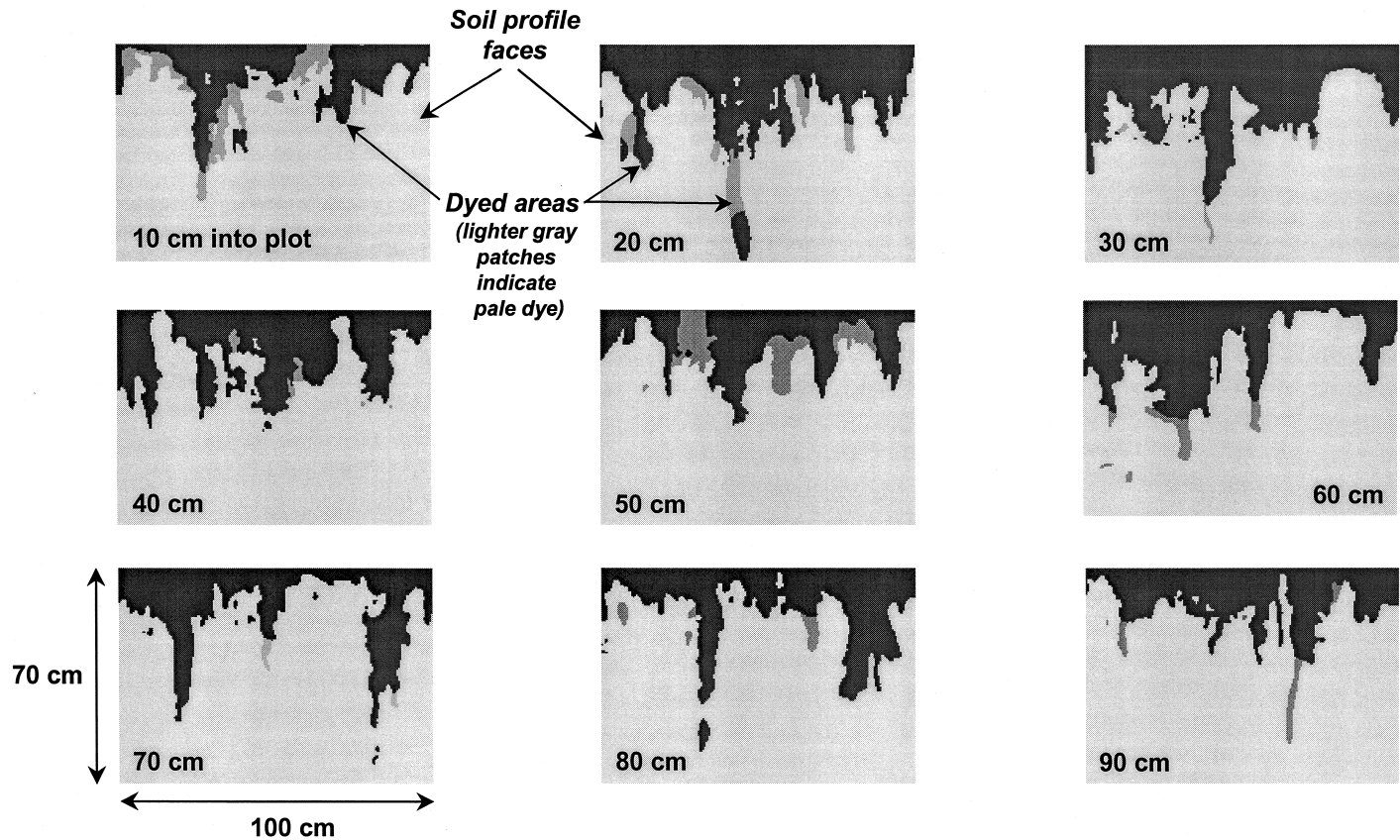


Fig. 2. Example of typical dye patterns seen under FLOOD application. These are digitized images, to scale, taken from color slides of excavated profiles (see text). Black areas are dyed areas, the medium gray areas are areas that were only slightly colored by the dye (i.e., pale), and the light gray background shows the 1 m wide profile face. The top of each profile shown is 25 cm below the soil surface (approximately at the Ap–Bt horizon boundary) extending to 95 cm below the soil surface. These nine profiles were taken sequentially at 10 cm increments beginning 10 cm into the plot.

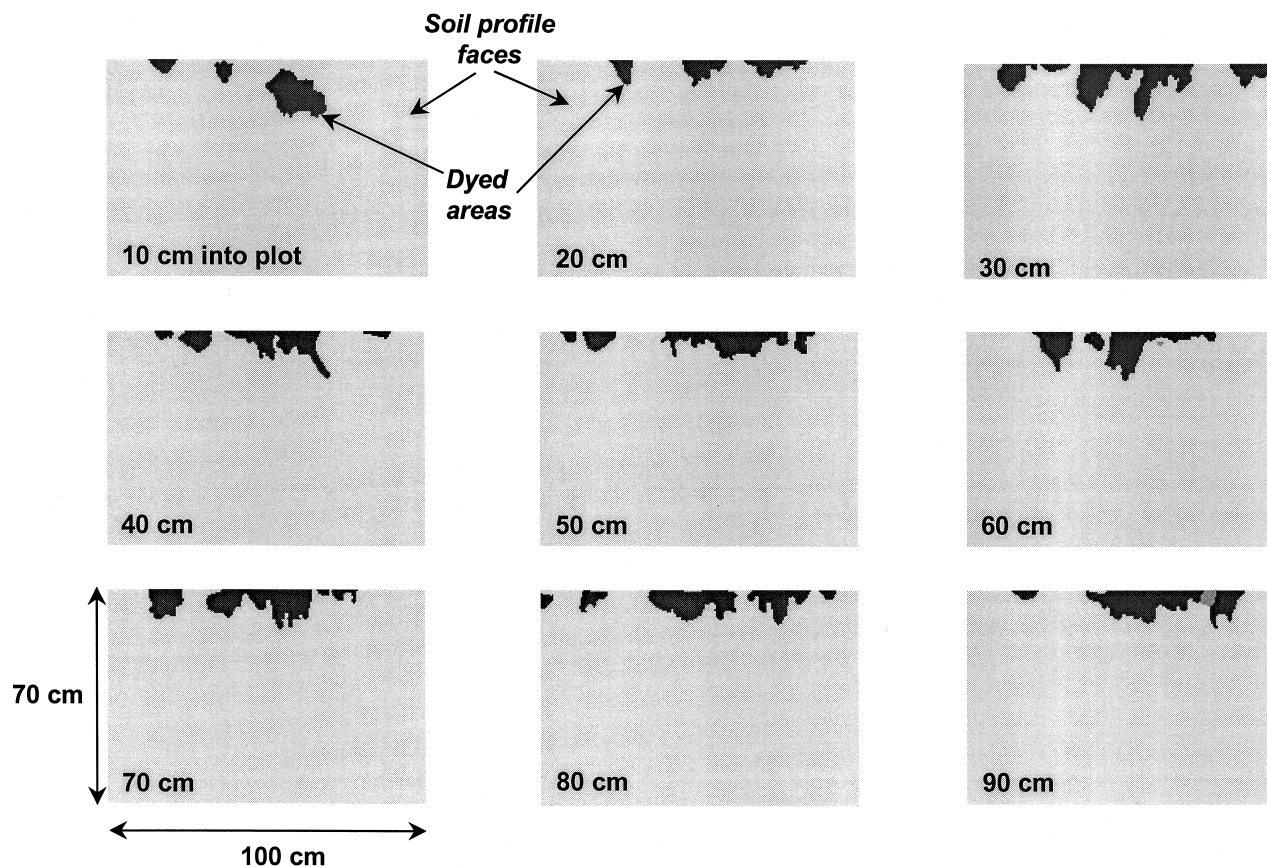


Fig. 3. Example of typical dye patterns seen under SPRINKLER application. These are digitized images, to scale, taken from color slides of excavated profiles in the first replicate of the SPRINKLER-HIGH, WET treatment. Black areas are dyed areas, the medium gray areas are areas that were only slightly colored by the dye (i.e., pale), and the light gray background shows the 1 m wide profile face. The top of each profile shown is 25 cm below the soil surface (approximately at the Ap–Bt horizon boundary) extending to 95 cm below the soil surface. These nine profiles were taken sequentially at 10 cm increments beginning 10 cm into the plot.

uniform dye fronts compared to the FLOOD plots—with relatively short PFPs that generally extended only a few centimeters below the dye front. The shorter PFPs that remained within the Bt horizon or upper part of the 2BC were less cylindrical in shape and often had more lateral components. At all application rates and initial moisture conditions, the color boundary between dyed and undyed zones was usually very sharp, suggesting little lateral spreading of the dye. There was no statistical difference in field moisture content or bulk density between dyed and no-dye regions in a given profile.

In several plots, especially under drier conditions, the dye patterns were extremely complicated and had intricate patterns of dyed and undyed soil (Fig. 4). This intricate pattern was observed particularly in the ‘no-alfalfa’ plots. There were no easily observable soil morphologic features that could be clearly related to these intricate patterns, therefore the mechanism(s) creating these patterns were not discerned. It is possible that the intricate patterns may have resulted from small features that were not easily observable—i.e., weak soil structure or fine root channels (< 0.5 mm). Another possibility is that these

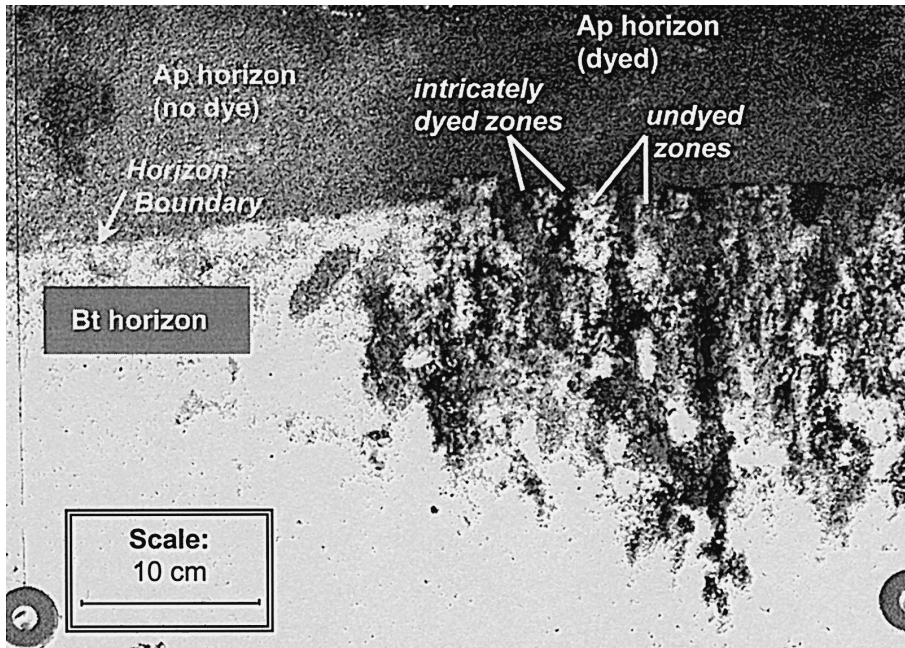


Fig. 4. Example of the intricate patterns of dyed and undyed zones within the soil as observed in many profiles. Within the Bt horizon, the black areas are dyed while the white areas are not dyed. The top of the image is at the soil surface, and the image is approximately 50 cm wide. This black and white image was created by enhancing a color image scanned from a color slide. The major steps taken in the enhancement were to split the color image into three color bands (red, green, and blue), and then divide the values of the pixels in the blue image by the values in the red image.

patterns may have developed due to relatively dry initial water contents (generally $< 0.10 \text{ cm}^3 \text{ cm}^{-3}$) in these plots, which caused lower hydraulic conductivity (limiting lateral spreading) or caused water repellency effects.

3.3. Patterns in dye coverage of profile faces

Patterns in the amount of dye coverage (fractional area dyed) at each depth for each treatment are shown in Fig. 5. Each line represents the mean of the nine profile faces for a given plot. Initial soil wetness did not appear to affect dye coverage (dyed area). However, initial soil wetness did appear to influence the *depth* of dye penetration in some cases. In the alfalfa FLOOD plots, deepest dye penetration did occur in initially WET soils in both replicates. Flury et al. (1994) also observed that initially wetter soils resulted in deeper dye penetration in most cases. Application rate did affect dye coverage. The curves in Fig. 5 shows that a greater proportion of the total profile face was dyed under FLOOD application, compared to the SPRINKLER application. There was little differ-

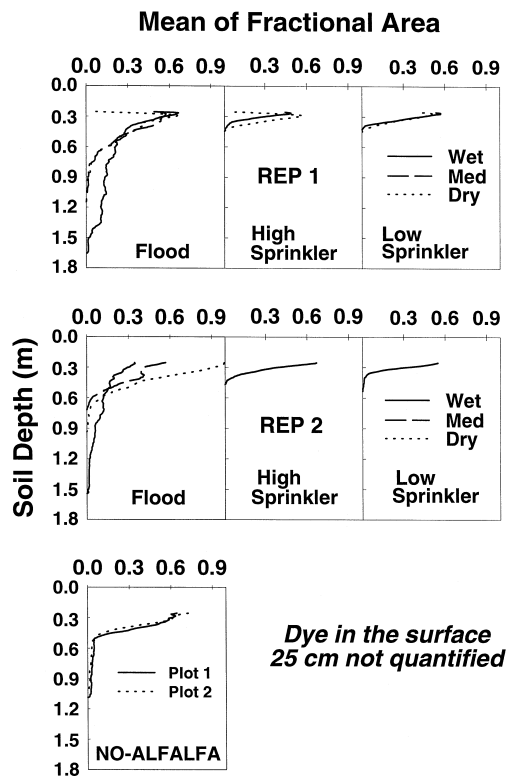


Fig. 5. Patterns in fractional dye coverage of the profile with depth. Each line represents the mean of the fractional area dyed at each depth for the nine profiles excavated for each plot. Dye patterns quantified represent the area below 25 cm depth as explained in the text.

ence in the proportion of dye coverage between the various SPRINKLER treatments.

Fig. 5 also shows that there was deeper movement of the dye under the FLOOD plots compared to either SPRINKLER treatment, and there was little difference in the depth of dye penetration between the two SPRINKLER treatments. This observation might be interpreted to mean that *water* and *other solutes* will also move more deeply under FLOOD applications compared to SPRINKLER applications. However, a subsequent laboratory study (Perillo et al., 1998) found that there was substantially greater dye retardation at slower application rates corresponding to the SPRINKLER-HIGH treatment than under the FLOOD application treatment, indicating that depth of dye movement is not a good surrogate for depth of water movement. Therefore, we hesitate to extrapolate the depth of dye penetration to the depth of water or other solute movement in similar situations.

High coefficient of variation (CV) among the 9 profiles in each plot for the mean amount of dye coverage (Fig. 6) show that there was extensive variation in

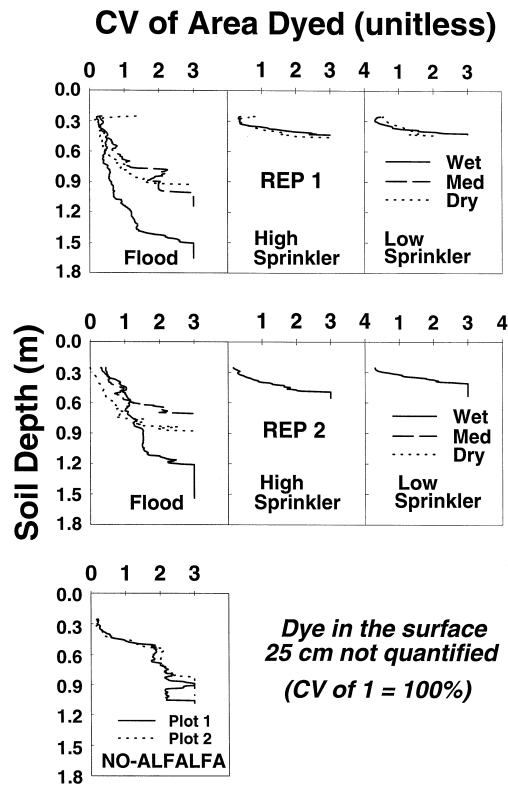


Fig. 6. Patterns of variability in the fractional dye coverage of the profile with depth, expressed as the Coefficient of Variation (CV) of the nine profile faces of each plot. (Calculated as: $CV = \text{standard deviation of dye coverage} / \text{mean dye coverage}$, where CV of 1 = 100%). Dye patterns in the surface 25 cm were not quantified (see text).

dye coverage at a given depth in a given plot. At depths of about 50 cm, CV values approached about 1.0 (100%) for most treatments, and about 3.0 (300%) near the bottom of the dyed zone of all treatments. This high variability in dye coverage demonstrates a large spatial variability in the transport of this adsorbed tracer within a single square meter. Other researchers have come to similar conclusions (that is, high spatial variability in solute concentrations being the result of preferential transport) working with other adsorbed chemicals such as pesticides at somewhat larger scales (Ghodrati and Jury, 1992, Cogger et al., 1998). The high variability also highlights the limitations of extrapolating preferential flow patterns from a single profile face to a larger area.

The number of PFPs present at a given depth is one index of the relative prevalence of preferential flow under the various treatments (Fig. 7), as it indicates how many separate flow paths were present at each depth. PFPs that extended below 55 cm were considered ‘deep PFPs’ since 55 cm is the depth the dye front would have moved if it had moved uniformly without dispersion or retardation due to chemical interaction with the soil. Since we know that this dye is retarded compared to the wetting front, especially under slow application

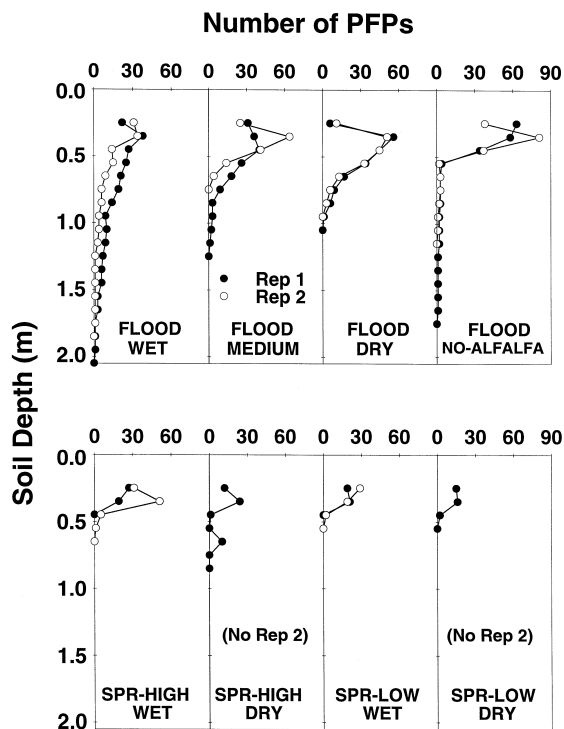


Fig. 7. Pattern in the number of individual preferential flow paths (PFPs) crossing a given depth. Each line represents the sum of the number of PFPs for the entire plot (i.e., the total for the nine profile faces in a given plot). SPR = SPRINKLER treatment.

rates (Perillo et al., 1998), the ‘deep PFPs’ were considered areas of significant preferential transport that had relatively rapid flow conditions. We observed many PFPs extending below 55 cm—particularly under FLOOD conditions. In fact, all plots receiving FLOOD application—regardless of initial wetness—contained at least one PFP extending below 1 m depth (Fig. 7). Relatively fewer PFPs were observed below 55 cm in the SPRINKLER treatments, and these did not extend as deeply as those in the FLOOD plots.

3.4. *Preferential flow path initiators*

Observations made during excavation of each profile suggested that once the dye solution entered the Bt horizon (regardless of mechanism), it continued to move vertically down with little or no lateral deviation, regardless of the material it passed through. For example, in at least six profiles dye movement occurred along a root or decayed root through the Bt horizon, and continued to flow vertically even if the root bent laterally (e.g., along the bottom of the Bt horizon). Hence, we began to think of these features as PFP *initiators*, since flow established by one of these features continued to move vertically (i.e., by gravity) without regard to other morphologic features. Several different types of PFP initiators were discerned (Table 3), including: roots and decayed roots; characteristics of the Ap–Bt horizon boundary; and less frequently, open burrows and krotovinas (filled burrows, originally excavated by insects or rodents).

3.4.1. *Roots*

Roots and decaying roots consisting of lignified vascular tissue appeared to be the most important PFP-initiators for the very deep PFPs in the alfalfa FLOOD plots. An example of the relationship between a decayed root and a PFP is shown in Fig. 8. Other researchers (Zins et al., 1991; Li and Ghodrati, 1994), using soil columns, have also found that alfalfa roots contributed to preferential transport of chemicals through soil.

The importance of roots as PFP initiators became increasingly apparent over the course of excavating the numerous profile faces. The category of ‘undifferentiated’ roots (Table 3) includes both decayed and live roots. With hindsight, decayed roots were the most important type but the relative percentage was not recorded for many of the first plots excavated. It was not necessarily all roots (decayed or live) that were important, as there were often roots that were unstained and not associated with PFPs. The pattern we observed was that deep PFPs were associated with decayed roots that crossed the Ap–Bt horizon. The decayed vascular tissue tended to keep a partially open channel, and thus provided a stable inlet for dye solution to enter the lower-permeability Bt horizon. In the decayed root channels we often found smaller live roots following these channels, likely due to the ease of penetration in these channels compared to the soil matrix itself. It seems probable that this pattern acts as a

Table 3

Prevalence of preferential flow path (PFP) initiators observed in the field. The number in a column indicates the number of soil profiles that had one or more PFPs which appeared to have been initiated by the corresponding type of feature. See text for explanation of PFP initiators. There were nine profiles excavated in each plot. Note that more than one type of initiator may be found in the same profile

Applic. rate	Initial moisture	Rep	Number of profiles								Other	Total
			Roots			Fauna		Ap–Bt Boundary				
			Undiff.	Decayed	Bent	Krotovina	Open burrow	Mixed	Sloping			
FLOOD	WET	1	7	1	1	2	1	5	0	1	18	
	WET	2	3	0	0	0	0	5	0	0	8	
	MED	1	7	0	0	1	0	2	0	2	12	
	MED	2	5	1	1	0	0	2	0	0	9	
	DRY	1	5	5	4	0	0	1	0	0	15	
	DRY	2	3	8	0	0	0	0	0	0	11	
SPRINKLER-HIGH	WET	1	5	1	0	1	0	4	0	1	12	
	WET	2	6	5	0	2	0	0	0	0	13	
	DRY	1	6	1	0	0	0	1	0	0	8	
SPRINKLER-LOW	WET	1	2	1	0	1	0	2	0	2	8	
	WET	2	2	0	0	0	0	2	1	2	7	
	DRY	1	1	0	0	2	0	1	0	2	6	
NO-ALFALFA Plots		1	3	2	0	0	3	3	3	0	14	
		2	6	0	0	2	1	7	0	0	16	
	Total		61	25	6	11	5	35	4	10		

UnDiff. = ‘undifferentiated’, i.e., no record made of whether roots were live or decayed. Dec. = Decayed root with lignified vascular tissue remaining. Bent = Situations where dye moved vertically in spite of lateral bend in the root. Krotovina = filled-in animal burrow.

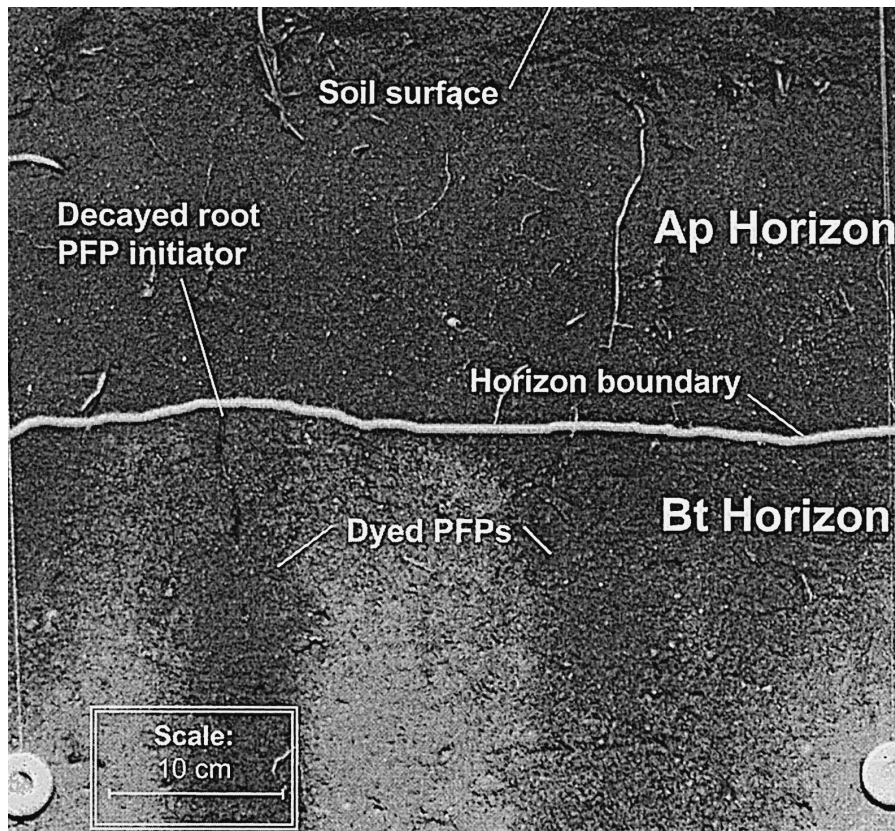


Fig. 8. Example of the relationship between a PFP initiator and the PFP itself as seen in the field. The darker stained 'PFP initiator' feature shown here is lignified vascular tissue that is stained with dye. This black and white image has been converted from a scanned color slide. Within the Bt horizon the darker areas are dyed, and the lighter areas are undyed. Most of the Ap horizon in this image is dyed, but these areas can not be differentiated in this image.

positive feed-back loop and that once this type of PFP-initiator is in place, it may continue to act as an initiator of preferential flow paths in the same location for many years.

In the NO-ALFALFA plots, roots and decayed roots were not important for initiating PFPs. This may be due to lack of alfalfa roots specifically, since numerous other taproots from a variety of herbaceous species were observed in these plots. Another explanation of the limited role played by roots in these plots is that most of the infiltrating dye solution was immediately diverted into open burrows (discussed below).

3.4.2. Horizon boundary characteristics

Two types of Ap–Bt horizon boundary characteristics were associated with dye PFPs: 1) boundary distinctness, and 2) boundary 'topography' (e.g., wavy,

sloping, etc). Both abrupt and gradual boundaries between these horizons were observed in all plots. The abrupt boundaries were created by plowing. The gradual boundaries consisted of ‘mixed’ Ap and Bt material. The term ‘mixed’ as used here includes two different cases. The first is where A horizon inclusions occurred in the upper Bt horizon, for example due to insect or root activity, and would be described as A/Bt if they had larger lateral extent. The second case is where the A horizon extended more deeply than the plow zone, which would be described as a transition ABt or as A2. Practically speaking, these cases represent two ends of a spectrum, and in the field they were not always distinguishable (hence grouped together). In general, the lateral extent of ‘mixed’ boundary zones was 4 to 30 cm.

‘Mixed’ Ap–Bt boundaries appeared to be prevalent PFP initiators (Table 3), as they were associated with PFPs in 35 of the 126 profiles evaluated. A frequently observed pattern was that there would be an accumulation of dye at abrupt boundaries and deeper movement of dye in the areas where the boundary was mixed. Deeper penetration in the ‘mixed’ zones might have been due to physical processes (higher permeability in the mixed zone compared to the Bt horizon), or it might have been due to low chemical adsorption onto soil in the mixed zones since the K_d value for the dye in the Bt horizon was found to be 5 times greater than that of the Ap horizon (Perillo et al., 1998). Sloping Ap–Bt horizon boundaries were not common since plowing was to a uniform depth. However in some cases small ‘hills’ in the boundary, a few centimeters tall and wide, did occur (Fig. 9). In these cases it appeared that the dye ran along the ‘sides’ of the hill, leading to penetration of the dye into the Bt horizon at the ‘toeslope’ of the hill. These patterns are similar, though smaller in scale, to the deeper movement of solutes in the tongue portions of an irregular B–C horizon boundary observed in a sandy soil by van Wesenbeeck and Kachanoski (1994).

PFPs associated with changes in horizon boundary characteristics were generally shorter than those associated with roots, especially under FLOOD conditions. The difference in depth of dye penetration from these two types of initiators might be due to differences in the amount of dye solution moving in these pathways. It might also be due to differences in the rate of solution movement, as discussed earlier for the generally deeper penetration under FLOOD conditions compared to SPRINKLER application. The dye solution in PFPs associated with horizon boundaries may have moved as matrix flow, while dye solution associated with roots may have included relatively rapid flow through partially open channels.

3.4.3. Open burrows and krotovinas

Open burrows were uncommon and were not associated with PFPs in any of the alfalfa plots. They were also rare in the no-alfalfa plots—with only one open burrow observed in each plot. However each of these open burrows (about 2 mm diameter, likely made by ants) appeared to be responsible for all but one of

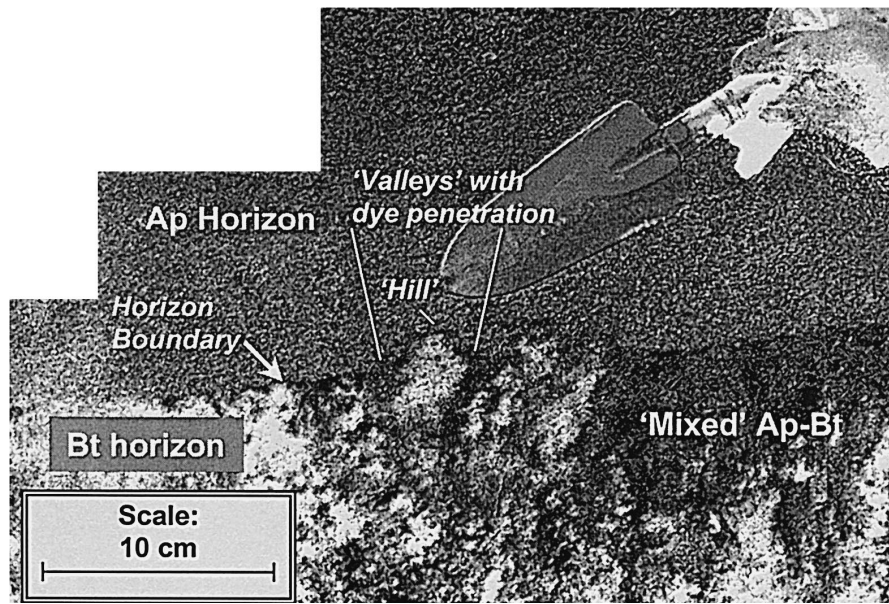


Fig. 9. Example of how characteristics of the horizon boundary affected patterns in dye movement. Within the Bt horizon, the black areas are dyed while the white areas are not dyed. The dye appears to have penetrated more deeply into the 'valley' areas of the horizon boundary compared to the 'hill' areas. Also there was deeper dye penetration into the portions of the profile that have 'mixed' Ap–Bt boundary compared to abrupt boundary areas. The top of the image is at the soil surface, and the image is approximately 50 m wide. This black and white image was created by enhancing a color image (scanned from a color slide). The major steps taken in the enhancement were to split the color image into three color bands (red, green, and blue), and then divide the values of the pixels in the blue image by the values in the red image.

the deep PFPs observed in these plots. We suspect that these pathways took in most of the applied dye solution, such that little dye solution was available for deep penetration in other parts of the plot. In contrast, Wang et al. (1996) found that ant burrows had little effect on water flow in a Wisconsin sand, mainly because they tended to close during irrigation events.

In both cases, the open burrows curved laterally and opened into larger 'chambers' within the Bt horizon. The PFPs continued vertically below the bottom of these chambers. In one plot, the center of the PFP directly below the chamber contained a vertical line of undyed sand (about 5 mm diameter) and organic material which appeared to be highly-decayed root material. The sand did not appear to be connected to the chamber. This sand was loose and likely had greater porosity and larger pores than adjacent dyed areas. One explanation for lack of dye in this material is that the dye solution in the surrounding dyed areas never had great enough hydraulic head to overcome the water entry pressure of this loosely-packed undyed zone. Hydrophobicity in this material is another possible explanation.

Large (> 2 cm diameter) krotovinas were observed in nearly every profile excavated, however most were not associated with PFPs. In two profiles (one FLOOD-MEDIUM and one NO-ALFALFA treatment), PFPs longer than 10 cm were associated with relatively large diameter krotovinas filled with light colored sand from the C horizon. All of the other krotovinas that were associated with PFPs were < 2 cm diameter, < 10 cm long, and connected to 'mixed' regions in the A–Bt horizon boundary—such that they might be considered an extension of the 'mixed' Ap/Bt horizon boundary.

3.4.4. Other initiators

No 'funnel' flow along depositional beds was observed—the most similar observation was flow along sloping horizon boundaries. However undisturbed parent material was rarely observed within 1.5 m of the soil surface in these plots. When depositional beds were present, they were deeper than the terminus of the PFPs, or the PFPs were not present in the portion of the profiles containing depositional patterns. A few PFPs were observed that might be attributed to unstable flow. These occurred under SPRINKLER conditions as short, finger-like protrusions of dye at the Bt–2BC boundary, extending below the dye front. Some of these protrusions could be attributed to roots or krotovinas, however for a few, no such initiating features were observed. Hydrophobicity (e.g., Ritsema and Dekker, 1994) is another possible PFP initiator since it is frequently observed in sandy soils, however we did not evaluate the hydrophobic characteristics of this soil.

4. Conclusions

The results of this study support Jury and Roth's (1990) conclusion that preferential flow (compared to movement in a uniform front) is probably the norm—as it certainly was the norm for the sandy loam over sand outwash soil we studied under a variety of application rate and initial wetness treatments. Our results also suggest the dominance of visible soil features such as roots, characteristics of horizon boundaries, and open burrows as probable initiators of preferential flow paths. Only a few (10 of 126) of the profiles excavated had preferential flow paths that were not associated with visible soil features.

Excavation of multiple profile faces within relatively small plots (1 m² each) discerned great variability—even between successive profiles—in the pattern, number, and depth of preferential flow paths. Our findings give substantial support to the speculation by several researchers that spatial variability patterns in soil pesticide concentrations is due to preferential flow.

One of our initial goals was to compare the relative length of preferential flow paths resulting from our application rate and initial soil wetness treatments, in order to make inferences about patterns in water moment. Subsequent

research, published elsewhere (Perillo et al., 1998), found that retardation of this dye in our soil was higher at a slow application rate compared to a faster application rate. Therefore, the variation in how deeply dye staining extended in this study depended not only on how deeply water moved, but also on the *rate* of movement. As a result, we emphasize that our findings of generally shallower PFPs under SPRINKLER conditions do not necessarily indicate lack of deeper water or nonadsorbing chemical (e.g., NO_3^-) movement under these conditions. Our findings may be somewhat correlated with relative PFP depth of other adsorbed chemicals (such as pesticides) under similar conditions, however this would need to be substantiated through further research.

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